CONTINUOUS SIGN LANGUAGE RECOGNITION USING HYBRID TRANSFORMER ARCHITECTURES

A Thesis
Presented to
The Academic Faculty

by

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Certificate

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Prajna P.N.
Supervisor

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Finally, I dedicate this work to my children, Viaansh and Avyaan, whose presence is my greatest motivation and joy.

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LIST OF SYMBOLS AND ABBREVIATIONS

List of Abbreviations

ASR	Automatic Speech Recognition
BLEU	Bilingual Evaluation Understudy score, commonly used in sequence generation quality assessment
BiLSTM	Bidirectional Long Short-Term Memory network
CNN	Convolutional Neural Network
CSLR	Continuous Sign Language Recognition
CTC	Connectionist Temporal Classification
DGS	Deutsche Gebardensprache (German Sign Language)
FPS	Frames Per Second
GUI	Graphical User Interface
HMM	Hidden Markov Model
I3D	Inflated 3D ConvNet for spatiotemporal feature extraction
PE	Positional Encoding
ReLU	Rectified Linear Unit activation function
ResNet	Residual Neural Network
RWTH-PHOENIX	RWTH-PHOENIX-Weather 2014 benchmark dataset
SLR	Sign Language Recognition
VAC	Visual Alignment Constraint
VRAM	Video Random Access Memory on GPU hardware
WER	Word Error Rate

List of Symbols

λ_{CE}	Weight for cross-entropy decoder loss in the joint objective
λ_{CTC}	Weight for CTC loss in the joint objective
d_{model}	Transformer hidden representation dimension
Q,K,V	Query, key, and value projections in self-attention
T	Sequence length in frames or timesteps
h	Number of attention heads
D	Feature dimension of latent representation
$\hat{Y}$	Predicted gloss sequence
Y	Ground-truth gloss sequence
$\mathscr{L}_{CTC}$	Connectionist Temporal Classification loss term
$\mathscr{L}_{CE}$	Cross-entropy loss term for autoregressive decoding
$\mathscr{L}_{hybrid}$	Total hybrid training objective combining CTC and CE terms

ABSTRACT

Continuous Sign Language Recognition (CSLR) is a challenging sequence learning problem because gloss boundaries are not explicitly available at frame level and visual patterns vary substantially across signers. This thesis presents a hybrid Connectionist Temporal Classification and attention transformer architecture for CSLR on the RWTH-PHOENIX-Weather 2014 dataset. The model uses a ResNet-18 based visual encoder and a transformer sequence module with two training objectives: Connectionist Temporal Classification for alignment learning and autoregressive cross-entropy for sequence modeling. In observed project runs, the hybrid objective showed more stable optimization behavior than pure Connectionist Temporal Classification baselines that exhibited blank-dominant trends. The implemented system achieves a best test Word Error Rate (WER) of 50.91% with practical training constraints on limited GPU memory. In addition, a bidirectional prototype is developed by integrating sign-to-text recognition and text-to-sign video retrieval in a desktop interface, demonstrating the feasibility of an assistive communication workflow.

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Chapter 1

INTRODUCTION

1.1 Background and Motivation

According to global public-health estimates, a large deaf and hard-of-hearing population depends on sign language as a primary mode of communication. Sign language is a complete linguistic system with its own grammar and discourse structure, and it communicates meaning through coordinated manual and non-manual cues. In practice, a persistent shortage of trained interpreters creates communication barriers across education, healthcare, governance, and routine social interaction.

Recent advances in deep learning have made automated sign language processing increasingly feasible. Early systems were mostly designed for isolated sign recognition in constrained environments. However, practical communication requires recognition of continuous sign streams in which sign boundaries are not explicitly marked. Continuous Sign Language Recognition (CSLR) therefore represents a substantially harder problem than isolated recognition.

1.2 The Core Technical Challenges

Transitioning from isolated gestures to continuous sentence-level recognition introduces multiple technical challenges:

- **Implicit Segmentation**: Continuous videos do not provide explicit frame-level sign boundaries, so alignment must be learned from weak supervision.
- **Co-articulation**: The appearance of a sign depends on surrounding signs, which changes visual trajectories and increases ambiguity.
- **Signer Variability**: Signing style, speed, body proportions, and camera conditions vary significantly across subjects.
- **Occlusion and Motion Blur**: Fast articulation and hand-face overlap often hide

discriminative visual details.

- **Limited Data**: Compared with speech benchmarks, publicly available CSLR datasets are relatively small, increasing overfitting risk for high-capacity models.

1.3 Problem Statement

A central bottleneck in CSLR is temporal alignment between long video sequences and shorter gloss sequences. Pure Connectionist Temporal Classification (CTC) formulations are attractive because they avoid explicit frame-level annotations, but they can become unstable in low-resource settings. Prior work reports CTC blank-dominant behavior (often called collapse), where the model over-predicts blank symbols and fails to produce informative gloss outputs.[1] In addition, long-range temporal dependencies remain difficult to model with purely alignment-driven objectives.

Therefore, this work targets a hybrid objective that combines alignment learning and sequence modeling, and extends the recognition pipeline toward a bidirectional assistive framework.

1.4 Objectives

The objectives of this thesis are:

1. Develop a hybrid CSLR architecture that combines a CNN visual backbone with transformer-based sequence modeling and joint CTC/cross-entropy supervision.

2. Evaluate the model on RWTH-PHOENIX-Weather 2014 under practical GPU memory constraints and report reproducible WER performance.

3. Design and implement a bidirectional assistive prototype integrating sign-to-text recognition and text-to-sign retrieval.

4. Demonstrate system usability through an interactive, multithreaded desktop interface.

1.5 Scope and Delimitations

The scope of this thesis is restricted to gloss-level continuous sign language recognition on the RWTH-PHOENIX-Weather 2014 benchmark and a practical bidirectional prototype for assistive communication. The following delimitations are explicitly acknowledged:

- The recognition model is evaluated on a single benchmark corpus with domain-specific content (weather broadcast context).
- The reverse communication module is retrieval-based rather than fully generative.
- Training is performed under limited GPU memory, which constrains batch size and temporal context length.
- The emphasis is on robust baseline construction, reproducible implementation, and practical usability rather than state-of-the-art leaderboard optimization.

1.6 Thesis Organization

The remainder of the thesis is organized as follows:

- Chapter 2 reviews prior work in CSLR, from classical pipelines to transformer and hybrid objective formulations.
- Chapter 3 summarizes the theoretical foundations of convolutional encoding, self-attention, and Connectionist Temporal Classification.
- Chapter 4 presents the proposed hybrid architecture and its training objective design.
- Chapter 5 details implementation decisions, augmentation policy, optimization protocol, and experimental results.
- Chapter 6 presents the bidirectional deployment prototype and system-level integration aspects.
- Chapter 7 concludes the thesis and outlines future research directions.

Chapter 2

LITERATURE REVIEW

2.1 Evolution of Sign Language Recognition

Automatic sign language recognition has evolved from sensor-based systems to modern vision-based deep learning pipelines. Early glove-based approaches offered controlled measurements but were intrusive and unsuitable for scalable deployment. Later vision-based methods using handcrafted features and classical classifiers reduced instrumentation requirements, but performance remained limited in unconstrained settings with variable lighting, backgrounds, and signer style.

In benchmark development, RWTH-PHOENIX-Weather established a large-vocabulary CSLR corpus for German Sign Language and became a standard evaluation setting for sequence models.[2] More recent datasets such as How2Sign expanded multimodal and multiview coverage for continuous sign-language research,[3] while OpenASL extended open-domain sign-language translation settings from online video sources.[4]

2.2 The Deep Learning Revolution

The introduction of deep convolutional neural networks (CNNs) reduced dependence on handcrafted descriptors by learning hierarchical spatial features directly from image data. A major step toward large-vocabulary CSLR was the CNN-HMM framework of Koller et al.,[5] which adapted sequence modeling concepts from automatic speech recognition. Although influential, HMM-based systems require carefully engineered state-transition assumptions and constrained alignment structure.

2.3 Connectionist Temporal Classification (CTC)

Connectionist Temporal Classification (CTC)[1] addressed explicit alignment requirements by summing over all valid alignment paths between frame sequences and target tokens. This made end-to-end training feasible without frame-level annotation. However, in long and noisy visual sequences, CTC can become blank-dominant, especially under limited data and weak contextual modeling. In practical use, CTC-only decoding is computationally attractive because it avoids autoregressive token-by-token decoding, but this efficiency can come with reduced contextual sequence modeling capacity.

2.4 Attention Mechanisms and Transformers

Transformer-based CSLR models improved long-range temporal modeling through self-attention. Early neural translation-oriented sign modeling work helped establish end-to-end sign-to-text learning settings.[6] Camgöz et al.[7] then demonstrated that joint recognition and translation pipelines can benefit from transformer encoders and decoders. In parallel, transformer-based sign recognition formulations were explored for direct recognition tasks,[8] while Zhu et al.[9] introduced visual alignment constraints to stabilize frame-to-token learning. Subsequent work including Hu et al.[10] further improved robustness by integrating stronger temporal correlation and hybrid supervision strategies. Relative to earlier pipelines, these approaches can model richer sequence dependencies, but they typically increase memory demand and training sensitivity.

From a broader vision perspective, transformer-based modeling progressed from image-transformer formulations[11] to dedicated video-transformer architectures such as TimeSformer,[12] with self-supervised video pretraining approaches like VideoMAE further improving data efficiency.[13]

These developments motivate the approach adopted in this thesis: a hybrid architecture that retains CTC-based alignment learning while adding autoregressive decoding supervision to improve optimization stability and sequence quality.

2.5 Comparative Perspective on Prior Methods

For clarity, Table 2.1 summarizes the methodological progression relevant to this thesis.

Table 2.1: Comparative Summary of Relevant CSLR Method Families

Method Family	Representative Works	Strengths	Common Limitations	Compute Trend
CNN-HMM pipelines	Koller et al.[5]	Strong alignment priors; interpretable sequence states	Rigid alignment assumptions; complex state design	Moderate
Pure CTC end-to-end models	Graves et al.[1]	No frame-level labels required; simple objective	Blank-dominant behavior in low-resource and long-sequence settings	Lower
Transformer-based CSLR	Camgöz et al.[7]	Better long-range temporal modeling; parallelizable sequence processing	Data-hungry; may still require alignment stabilizers	Higher
Hybrid / constrained objectives	Watanabe et al.,[14] Zhu et al.,[9] Hu et al.[10]	Improved optimization stability; better balance of alignment and language modeling	Additional hyperparameter sensitivity and training complexity	Higher

The compute/memory trend column is qualitative and reflects relative training and inference complexity reported in representative method families, not a controlled benchmark measured in this thesis.

2.6 Research Gap and Motivation for the Present Work

Despite substantial progress, two practical gaps remain for deployment-oriented systems under constrained resources. First, many high-performing methods assume larger compute budgets and broader hyperparameter sweeps than are feasible in constrained academic environments. Second, literature often emphasizes benchmark recognition accuracy but does not consistently extend the system into a usable bidirectional assistive workflow.

From a comparative viewpoint, prior hybrid architectures cited here focused primarily on improving recognition metrics, while bidirectional deployment support is less emphasized in these references. In many published pipelines, sign-to-gloss decoding is the primary target and reverse communication support is treated as a separate systems problem. This creates a practical gap between algorithmic performance and end-user assistive utility.

Accordingly, this thesis targets a balanced contribution: a stable hybrid CSLR baseline under constrained resources, accompanied by a practical bidirectional prototype that demonstrates integration beyond offline evaluation. This positioning is intentionally implementation-oriented and does not claim state-of-the-art benchmark performance.

2.7 Literature-Guided Experimental Decisions

The exploration strategy used in this thesis follows a literature-guided progression:

- Start with alignment-centric baselines from CTC literature.[1]
- Introduce hybrid objectives motivated by prior hybrid CTC-attention gains in sequence recognition.[14]
- Adopt transformer-based CSLR design patterns based on sign-language-specific evidence.[7]
- Evaluate alignment-constraint variants inspired by VAC.[9]
- Assess stronger video feature transfer choices such as I3D-style representations.[15]

This progression is reflected in the checkpoint evolution summarized later in the thesis appendix.

Chapter 3

THEORETICAL FOUNDATIONS

3.1 Introduction to Sequence Extraction

Continuous sign language recognition requires joint modeling of spatial information (within each frame) and temporal information (across frame sequences). The core theoretical components used in this thesis are convolutional feature extraction, self-attention based sequence modeling, and probabilistic alignment learning.

3.2 Deep Spatial Convolution and Residual Links

The visual feature extraction stage uses convolutional neural networks (CNNs). A standard convolution applies learnable kernels K over input image channels I:

$$(I * K)(i, j) = \sum_{m} \sum_{n} I(i-m, j-n) K(m, n) \tag{3.1}$$

Deep stacks of convolutional layers learn hierarchical representations, progressing from low-level edges to higher-level semantic patterns. In very deep networks, optimization may degrade due to vanishing gradients. Residual learning alleviates this by introducing identity shortcuts so each block learns a residual mapping $F(x)$:

$$H(x) = F(x) + x \tag{3.2}$$

This formulation improves gradient flow and enables stable training of deeper visual encoders. In this work, a ResNet-18 backbone is used to produce compact frame-level feature vectors.

3.3 The Transformer Attention Architecture

Temporal context is modeled using transformer layers rather than recurrent networks. Self-attention enables direct interaction between distant time steps and supports efficient parallel computation.

For an input sequence projected into Query (Q), Key (K), and Value (V) spaces, attention is computed as:

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3.3}$$

The scaling factor $\sqrt{d_k}$ stabilizes softmax values for high-dimensional embeddings. Since attention alone is permutation-invariant, positional encoding is added to inject temporal order:

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}}) \tag{3.4}$$

$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}}) \tag{3.5}$$

3.4 Connectionist Temporal Classification (CTC)

Alignment between long frame sequences and shorter gloss sequences is handled using Connectionist Temporal Classification (CTC). CTC computes sequence likelihood by marginalizing over all valid alignment paths π that map inputs to target sequence Y, including blank symbol $\emptyset$:

$$L_{CTC} = -\ln P(Y|X) = -\ln \sum_{\pi \in \Phi(Y)} \prod_{t=1}^{T} y^t_{\pi_t} \tag{3.6}$$

This objective enables end-to-end alignment learning without frame-level annotations.

3.5 Joint Objective Interpretation

The hybrid learning objective combines complementary inductive biases:

- The CTC term encourages monotonic frame-to-gloss alignment under weak supervision.

- The decoder cross-entropy term encourages sequence consistency and contextual token dependency modeling.

In practical terms, the objective can be interpreted as a weighted multi-task optimization problem where one branch promotes alignment robustness and the other branch promotes sequence-level fluency. Let

$$L = \lambda_{CTC} L_{CTC} + \lambda_{CE} L_{CE}, \quad \lambda_{CTC} + \lambda_{CE} = 1. \tag{3.7}$$

Increasing λ_{CTC} emphasizes alignment supervision, while increasing λ_{CE} emphasizes autoregressive sequence modeling. The selected balance in this thesis is chosen to reduce blank-dominant behavior while retaining alignment signal.

3.6 Evaluation Metric Foundation

The primary evaluation metric is Word Error Rate (WER), computed through edit operations between reference and predicted gloss sequences:

$$WER = \frac{S + D + I}{N}, \tag{3.8}$$

where S is substitutions, D is deletions, I is insertions, and N is the number of reference tokens.

WER is appropriate for CSLR because it directly measures sequence-level recognition quality and penalizes both missing and spurious gloss predictions. In addition to aggregate WER, monitoring blank-token behavior during training provides useful diagnostic evidence for identifying unstable optimization regimes.

Chapter 4

PROPOSED HYBRID CSLR MODEL

4.1 Introduction to the Hybrid Approach

This chapter presents the architecture implemented for continuous sign language recognition in this thesis. The main design objective is to improve optimization stability in low-resource CSLR by combining alignment-based and autoregressive supervision in a single end-to-end model.

4.2 Unified ResNet-18 Extraction Pipeline

The input is a video tensor of shape $T \times 3 \times 260 \times 210$. A ResNet-18 backbone pretrained on ImageNet is used to extract frame-level spatial representations. The final classification layer of ResNet is removed, and pooled feature vectors are projected to a shared model dimension $D = 512$.

To improve transfer stability and reduce overfitting risk on limited data, early backbone layers are frozen during training using parameter-level gradient control.

4.3 Shared Transformer Encoder Parameters

Frame embeddings are passed to a transformer encoder with $N = 6$ layers and $h = 8$ attention heads. Each encoder block contains multi-head self-attention and a feed-forward network with hidden size 2048, GELU activation, residual connections, and dropout regularization. Positional encoding is added before encoder processing to preserve temporal order.

4.4 Dual Head Implementation and Joint CTC-CE Loss Arrays

The encoder output is consumed by two task heads trained jointly:

4.4.1 Connectionist Temporal Inference Node

The CTC head applies a linear projection over encoder states to predict gloss vocabulary probabilities at each time step. The output space includes the gloss vocabulary and one blank token. This branch contributes the CTC alignment loss L_{CTC}.

4.4.2 Autoregressive Attention Decoder Output

A transformer decoder with 3 layers performs autoregressive gloss prediction using causal masking and cross-attention over encoder memory. This branch contributes sequence-level cross-entropy loss L_{CE} with label smoothing.

The total training objective is:

$$L_{total} = \lambda_{CTC} \times L_{CTC} + \lambda_{CE} \times L_{CE} \tag{4.1}$$

Figure 4.1 illustrates the complete hybrid CSLR pipeline with the shared encoder and dual-objective heads used in this thesis.

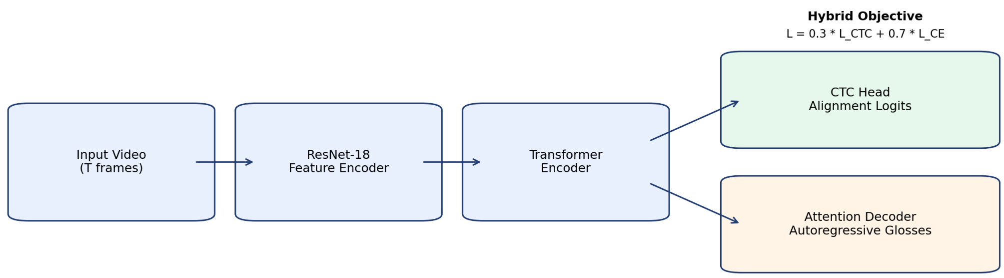

Figure 4.1: Hybrid CSLR pipeline with shared encoder and dual objective heads.

In this implementation, $\lambda_{CTC} = 0.3$ and $\lambda_{CE} = 0.7$. This weighting was selected to retain alignment supervision while prioritizing decoder-guided sequence modeling, which improves stability and reduces blank-dominant behavior compared with pure CTC optimization.

4.5 End-to-End Tensor Flow

Table 4.1 summarizes the principal tensor transitions in the forward pipeline.

Table 4.1: Main Tensor Flow Through the Hybrid Model

Stage	Input Shape	Output Shape
Video input	$(B, T, 3, 260, 210)$	$(B, T, 3, 260, 210)$
ResNet-18 backbone	$(B, T, 3, 260, 210)$	$(B, T, 512)$
Positional + encoder stack	$(B, T, 512)$	$(B, T, 512)$
CTC projection head	$(B, T, 512)$	(T, B, V)
Autoregressive decoder	$(B, T, 512)$ with shifted targets	(B, L, V)

4.6 Training-Time and Inference-Time Behavior

4.6.1 Training-Time Behavior

During training, the model receives full video clips and teacher-forced decoder targets. Both losses are computed in the same forward pass. This shared-encoder design improves parameter efficiency and allows gradients from alignment and sequence modeling objectives to jointly shape encoder representations.

4.6.2 Inference-Time Behavior

At inference, the decoder generates tokens autoregressively until an end token or maximum length is reached. In parallel, CTC outputs can still be inspected as an auxiliary signal for diagnosing alignment quality and blank-dominant trends.

4.7 Design Rationale Under Resource Constraints

The architecture choices in this thesis are guided by memory-aware deployment constraints:

- ResNet-18 is selected as a practical compromise between representational strength and computational footprint.

- Temporal windowing is capped to maintain feasible batch-level training on available GPU memory.
- Joint objective learning is preferred over larger model scaling, since objective design yields more stable gains under constrained resources.

Chapter 5

EXPERIMENTAL SETUP AND RESULTS

5.1 Dataset and Experimental Setup

All experiments were conducted on the RWTH-PHOENIX-Weather 2014 dataset, which is a standard benchmark for continuous sign language recognition. The dataset contains gloss-level annotations and is widely used for evaluating sequence-to-sequence CSLR models under realistic signer and sentence variability.

Table 5.1: RWTH-PHOENIX-Weather 2014 Subset Configuration

Dataset Property	**Value**
Sign language	Deutsche Gebärdensprache (DGS)
Domain	Meteorological forecasts
Gloss vocabulary size	1,236
Training sequences	5,672
Development sequences	540
Test sequences	629

The training environment imposed practical memory constraints. To maintain stable training on available GPU resources, the input clip length was limited to $T = 64$ frames and the mini-batch size was set to 4. These limits were consistently used for all reported experiments.

5.2 Data Augmentation Strategy

To reduce overfitting on the limited training set, a video-level augmentation pipeline was implemented in the VideoAugmentation module. The same sampled parameters are applied consistently across frames in a clip to preserve temporal coherence.

The augmentation operations include:

- Random spatial crop with scale in $[0.85, 1.0]$,

- Random rotation within $\pm 8°$,
- Color jitter (brightness, contrast, saturation) within ± 0.15,
- Mild Gaussian noise,
- Temporal masking with 15% probability and 8% mask ratio.

Horizontal flipping was intentionally excluded because mirror transforms can alter sign meaning and therefore introduce incorrect supervisory signals.

5.3 Temporal Sampling

A temporal sampling module was used to normalize variable-length sequences to fixed input length. During training, speed perturbation was applied in the range $(0.8\times, 1.2\times)$ to improve robustness against signer-dependent pace variation. During validation and testing, deterministic sampling was used for consistent evaluation.

5.4 Optimization and Training Protocol

The model was trained using AdamW with weight decay 0.05.[16] Differential learning rates were used: 1×10^{-5} for the pretrained CNN backbone and 1×10^{-4} for the remaining network parameters. A 5-epoch linear warmup followed by cosine decay was used over a maximum of 100 epochs, following warm-restart scheduling principles.[17] Gradient clipping with max norm 5.0 and early stopping with patience 15 were applied for stability. Decoder-side training used label smoothing, which is a standard regularization choice for sequence models.[18]

5.4.1 Compute Environment Note

Experiments were executed under constrained single-GPU settings. Repository utilities and training scripts record runtime environment details (for example, GPU name, available memory, and PyTorch version) at execution time (`check_system.py`, `train_improved.py`). However, complete hardware and software metadata were not consistently persisted for every historical run artifact; therefore this thesis reports configuration controls and logged metrics as the primary reproducibility evidence.

5.5 Evaluation Metric and Results

Performance was measured using Word Error Rate (WER), computed through Levenshtein distance between predicted and reference gloss sequences. The best development and test results obtained by the final hybrid configuration are reported in Table 5.7.

5.5.1 Experiment Evolution and Selection Rationale

Multiple model families were explored during the project lifecycle, including early transformer baselines, hybrid CTC-attention variants, ResNet-backed end-to-end branches, I3D-feature branches, an improved hybrid-training branch, and a VAC-augmented branch. The reasons for exploring these paths are grounded in published sequence-modeling and CSLR literature:

- CTC-based alignment modeling as a baseline,[1]
- Hybrid CTC-attention supervision for improved stability,[7, 14]
- Alignment-constrained objectives (VAC) for collapse mitigation,[9]
- Strong video feature backbones such as I3D for transfer learning in temporal vision tasks.[15]

Table 5.2 summarizes what each approach changed technically, why it was explored, and the observed qualitative outcome in this project.

Table 5.2: Approach-Level Motivation and Outcome Summary

Approach	What Changed Technically	Why It Was Tried	Observed Outcome
Transformer baseline	CTC-oriented transformer training without stronger hybrid stabilization	Establish a reference for alignment-first training and diagnose blank-collapse risk	Unstable convergence; high WER
Hybrid (CTC + Attention)	Joint objective with shared encoder and decoder supervision ($\lambda_{CTC}, \lambda_{CE}$)	Improve optimization stability and sequence quality reported in prior CSLR literature	Clear stability gain; selected as core direction
I3D feature branch	Replaced/augmented visual representation with I3D-style spatiotemporal features	Test whether stronger pretrained video features improve low-resource CSLR robustness	Did not outperform ResNet-18 hybrid path
Improved hybrid training	Enhanced temporal augmentation, gradient accumulation, and tuned training protocol	Reduce overfitting and improve generalization under limited GPU and data constraints	Better than early baselines; below best ResNet-18 branch
VAC family	Added visual alignment constraint losses (SeqCTC + ConvCTC + distillation) using one VAC implementation path	Mitigate alignment drift and collapse-like behavior through auxiliary alignment supervision	Mixed results across runs; best VAC run improved substantially

To improve experiment traceability, Table 5.3 maps each approach family to the corresponding training script and recorded evidence source used in this thesis.

Table 5.3: Approach Traceability to Repository Artifacts

Approach Family	Training Script	Recorded Evidence Source	Best Recorded WER
Transformer baseline	train_text2gloss.py	Archived checkpoint summary (Appendix A.1)	91.28%
Hybrid CTC+attention	train_hybrid.py	Final model reporting in this chapter	**50.91% (test)**
End-to-end branch	train_e2e.py	Archived checkpoint summary (Appendix A.1)	95.42%
Improved branch	train_improved.py	Archived checkpoint summary (Appendix A.1)	60.40%
I3D branch	train_i3d.py	Archived checkpoint summary (Appendix A.1)	94.78%
VAC family	train_vac.py	training_log_vac.jsonl and Appendix A.1	61.43%

Checkpoint evidence showed that several early branches remained unstable or underperformed, while the ResNet-18 branch yielded the strongest stable validation trajectory in this project context (best checkpoint WER $= 51.44\%$ at epoch 89 in `checkpoints/resnet18/best.pth`). In the appendix, VAC results were grouped as a single approach family because the implementation path was the same and differences across runs were attributable to training progression rather than distinct architecture definitions. A consolidated checkpoint-based summary was provided in Appendix A.1.

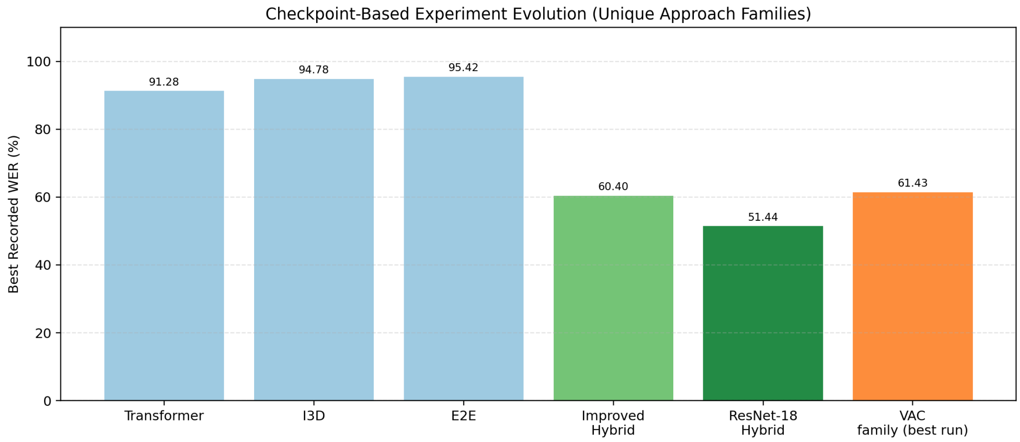

Figure 5.1: Checkpoint-based experiment evolution in terms of best recorded WER.

5.5.2 Comparative Observation with Preliminary Baselines

Preliminary experiments with a pure Connectionist Temporal Classification objective showed unstable convergence and blank-dominant predictions, which is consistent with the collapse behavior reported in CSLR literature. The hybrid objective improved training stability and produced substantially better sequence predictions. A summary of this comparison is provided in Table 5.4.

Table 5.4: Observed Baseline Comparison During Development

Model Variant	**Training Behavior**	**Outcome**
Pure CTC baseline	Unstable, blank-dominant outputs	Not suitable for final model
Hybrid CTC + attention	Stable optimization	Selected for final evaluation

5.5.3 Diagnostic Comparison Protocol

To check whether observed improvements were associated with objective design rather than incidental behavior, a diagnostic comparison protocol was used during development. The protocol compares:

- CTC-only training,
- Hybrid training with different loss-weight balances,
- Hybrid training with and without augmentation components.

For each run family, the following diagnostic signals were tracked alongside WER:

- blank-token ratio trend during training,
- validation-loss stability across epochs,
- qualitative sequence plausibility in decoded gloss outputs.

This protocol was useful for diagnosing collapse-like behavior early and for selecting practical operating regions for λ_{CTC} and λ_{CE}.

5.5.4 VAC Log Diagnostic Trend

In addition to checkpoint summaries, one complete structured training log (training_log_vac.jsonl) was available and used for objective diagnostic interpretation. This log corresponds to the VAC exploratory family and is included as supporting evidence rather than as the final model source.

Table 5.5: Diagnostic Snapshot from training_log_vac.jsonl

Epoch	Dev WER	Blank Ratio	Dev Loss
1	0.9592	0.9959	5.2855
20	0.6838	0.8938	3.4691
35	0.6143	0.8145	3.4617

Within this exploratory VAC log, both dev WER and blank ratio decreased substantially across epochs, which is consistent with improved training stability in that run family. These values are reported directly from logged records and are not used to replace the final selected-model test result.

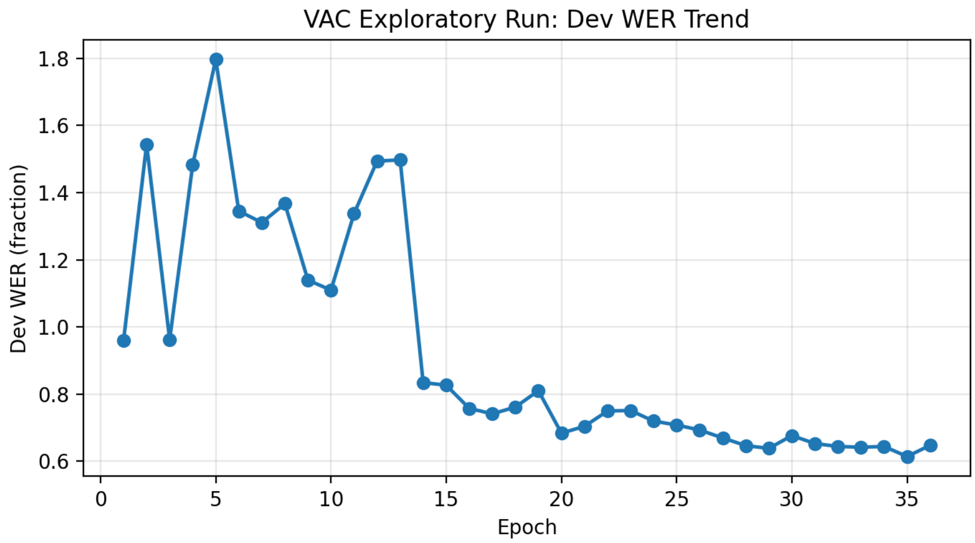

Figure 5.2: VAC exploratory run: development WER trend from training_log_vac.jsonl.

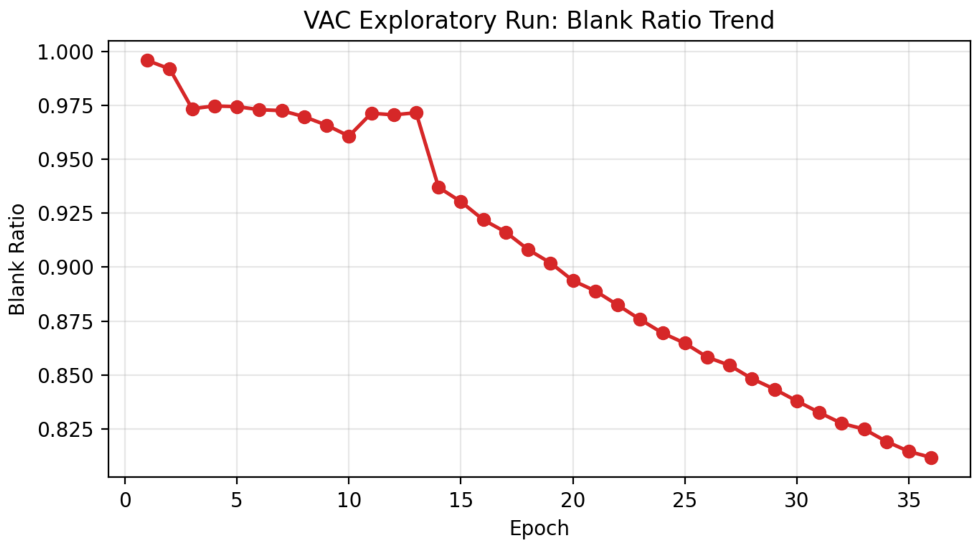

Figure 5.3: VAC exploratory run: blank-ratio trend from training_log_vac.jsonl.

5.5.5 Objective-Weight Evidence from Archived Runs

Table 5.6 summarizes objective-weight settings for runs where numerical evidence is retained in current repository artifacts.

Table 5.6: Available Objective-Weight Evidence from Archived Runs

Objective Setting	Weight Pair $(\lambda_{CTC}, \lambda_{CE})$	Recorded WER
CTC-oriented transformer baseline	(1.0, 0.0)	91.28% (archived branch checkpoint summary)
Selected hybrid final model	(0.3, 0.7)	**50.91%** (final reported test result)

This table reports only settings with retained numerical evidence in current artifacts. Intermediate balancing settings (for example $(0.5, 0.5)$) were explored during development but do not have a complete archived metric trail suitable for quantitative thesis reporting.

5.5.6 Observed Practical Trade-offs

The experiment trajectory also revealed practical trade-offs:

- Higher-capacity or constraint-augmented objectives require careful tuning; otherwise optimization can degrade quickly in low-resource settings.
- In observed project runs, objective balancing appeared more beneficial than increasing model capacity alone in the current compute regime.
- Data and temporal-context constraints remained dominant factors limiting final WER.

Table 5.7: System Performance Error Metrics

Analytical Component	Development Validation	Final Test Result
Optimal Word Error Rate	51.44%	**50.91%**
Blank token ratio (training)	12.3%	–
Minimum training loss	0.0006	–

The achieved test WER of 50.91% demonstrated that the proposed hybrid objective provided stable training behavior and a practically workable baseline under constrained hardware and limited-data conditions. In this thesis, 51.44% refers to the best development-side checkpoint from the ResNet-18 branch, while 50.91% is the final reported test result of the selected hybrid system.

5.6 Error Analysis Perspective

In sequence recognition, aggregate WER can mask distinct error sources. During qualitative review of decoded outputs, three dominant error types were observed:

- **Deletion-heavy errors**: missing glosses in fast-motion or occluded segments.
- **Substitution errors**: confusion between visually similar gloss motions.
- **Insertion errors**: occasional over-generation in transitional frames.

This pattern is consistent with continuous visual sequence ambiguity and motivates future work on stronger alignment constraints and longer temporal context modeling.

5.6.1 Observed Decode Example from Archived Checkpoints

Table 5.8 reports one real decode case extracted from archived checkpoint inference outputs using repository evaluation scripts (`scripts/extract_decode_examples.py`) on the test split.

Table 5.8: Example Decode Comparison (Archived Checkpoint Inference)

Type	**Sequence**
Ground truth	ABER FREUEN MORGEN SONNE SELTEN REGEN
CTC-oriented baseline output	(empty output)
Hybrid output	AUCH DANACH MEHR SONNE SONNE REGEN REGEN

In this sampled case, the hybrid output remains imperfect but recovers content tokens, while the baseline output collapses to an empty sequence. This observation is consistent with the broader collapse-mitigation narrative from log diagnostics.

5.6.2 Representative Decoding Error Cases

To make the error profile operationally useful, Table 5.9 summarizes schematic sequence-level failure patterns used to categorize qualitative observations.

Table 5.9: Schematic Sequence Error Pattern Categories

Error Type	Reference Pattern	Predicted Pattern	Likely Cause
Deletion-dominant	`g_1 g_2 g_3 g_4`	`g_1 g_3 g_4`	Intermediate transition missed under rapid motion
Substitution-dominant	`g_a g_b g_c`	`g_a g_d g_c`	Visual similarity and co-articulation overlap
Insertion-dominant	`g_x g_y g_z`	`g_x g_k g_y g_z`	Decoder over-generation near ambiguous transition frames
Boundary drift	`g_p g_q g_r`	`g_p g_q g_q g_r`	Alignment uncertainty around sign boundaries
Length truncation	`g_m g_n g_o g_u`	`g_m g_n g_o`	Early sequence termination in low-confidence regions

These schematic patterns were used as a qualitative categorization aid and were consistent with aggregate WER trends and blank-ratio diagnostics from unstable training phases. They are not presented as verbatim dataset-specific decode transcripts.

5.6.3 Error-Type-Oriented Improvement Mapping

The identified error classes directly mapped to technical interventions:

- **Deletion-heavy outputs**: improve temporal context coverage and alignment regularization.
- **Substitution-heavy outputs**: strengthen visual discriminability and signer-robust augmentation.
- **Insertion-heavy outputs**: tune decoding termination criteria and objective balance.

This mapping provided a principled path from qualitative inspection to measurable model-improvement experiments.

5.7 Reproducibility Notes

To improve reproducibility, the implementation records epoch-wise metrics in structured logs and saves both best and latest checkpoints. The dataset split policy, temporal clip length, batch size, optimizer class, and objective weights are kept fixed in the reported final runs. These controls reduce accidental variance and support consistent re-evaluation.

Chapter 6

BIDIRECTIONAL DEPLOYMENT PROTOTYPE

6.1 Motivation for Bidirectional Communication

Continuous sign language recognition addresses only one communication direction (sign-to-text). In practical assistive settings, a complete interaction loop requires both directions: recognition from sign to text and response generation from text to sign representation. For this reason, a bidirectional prototype was developed in addition to the core CSLR model.

6.2 Text-to-Sign Retrieval Design

Instead of direct neural video synthesis, this work uses a deterministic retrieval-based approach for reverse mapping. The design choice avoids visual artifacts and semantic instability that are common in unconstrained generative methods when training data is limited.

The reverse pipeline consists of three stages:

1. **Text normalization and tokenization**: Input text is standardized and segmented into units suitable for glossary lookup.
2. **Gloss-level mapping**: Tokens are mapped to available gloss/video entries through dictionary and rule-based matching.
3. **Video concatenation**: Retrieved sign clips are merged in sequence to produce a playable response stream.

6.3 Desktop GUI and Runtime Architecture

The prototype GUI was implemented using Python and Tkinter with OpenCV-based camera integration. The interface exposes:

- Live sign-to-text recognition from camera frames,
- Text input for reverse text-to-sign rendering,
- Playback controls for retrieved sign video output.

To maintain responsiveness, inference and I/O were separated using background threads. Camera capture, model inference, and UI rendering run asynchronously, preventing the interface from freezing during model execution or video concatenation.

6.4 Discussion

The bidirectional module is intended as a functional demonstration layer over the trained recognition model. While the reverse path is retrieval-based rather than fully generative, it is robust, deterministic, and suitable for validating interaction flow in constrained deployment settings. This integration demonstrates that the proposed CSLR system can be extended from benchmark evaluation to user-facing assistive prototypes.

6.5 System-Level Evaluation Considerations

The bidirectional prototype is evaluated primarily from a systems-integration perspective rather than from a standalone generative benchmark perspective. The main assessment dimensions are:

- functional correctness of sign-to-text and text-to-sign pathways,
- responsiveness of the GUI under concurrent camera capture and inference,
- robustness of retrieval-based playback under variable input phrases.

These criteria are aligned with assistive usability goals, where reliable interaction flow is often more critical than purely aesthetic synthesis quality. End-to-end latency and frame-rate benchmarks were not measured as separate quantitative deployment metrics in this work.

6.6 Practical Limitations

The current deployment design has known limitations:

- retrieval quality depends on available glossary coverage in indexed clips,
- phrase-level smoothness can degrade at clip boundaries,
- out-of-vocabulary text requires fallback handling.

Despite these constraints, the architecture provides a stable baseline for bidirectional assistive interaction and establishes a practical path for future enhancement.

Chapter 7

CONCLUSION AND FUTURE WORK

7.1 Concluding Remarks

This thesis presented a hybrid CTC-attention framework for continuous sign language recognition on the RWTH-PHOENIX-Weather 2014 benchmark. The work addressed a key practical challenge of low-resource CSLR training: instability of purely alignment-driven objectives under long and variable input sequences.

By combining CTC alignment supervision with autoregressive cross-entropy decoding, the proposed model showed stable optimization behavior in observed project runs and reduced collapse tendencies. Under constrained hardware settings, the final system attained a best test Word Error Rate of 50.91%, demonstrating the viability of the approach as a strong baseline for further improvements.

7.2 Scope and Validity Notes

The conclusions in this thesis should be interpreted within the following bounds:

- recognition results are reported for RWTH-PHOENIX-Weather 2014 and were not validated on additional CSLR corpora in this work,
- the reverse communication module is retrieval-based and was evaluated functionally rather than with a dedicated generative-quality benchmark,
- deployment latency and throughput were not reported as standalone quantitative metrics.

Within these limits, the reported findings are intended as evidence for a reproducible engineering baseline under constrained compute conditions.

7.3 Contribution Summary

The major contributions of this thesis are:

- Design and implementation of a hybrid ResNet-transformer CSLR model with dual CTC and attention objectives.
- A practical training protocol including augmentation, warmup-cosine scheduling, label smoothing, and gradient clipping for low-resource robustness.
- Empirical validation of model performance on RWTH-PHOENIX-Weather 2014 with reported development and test WER metrics.
- Development of a bidirectional prototype interface integrating sign-to-text recognition and text-to-sign retrieval for assistive communication flow.

In addition to final-model reporting, this thesis documents a full experiment-evolution trajectory across multiple model families and objective variants. This increases transparency of research decisions and supports reproducibility.

7.4 Objective-Wise Executive Conclusion

The objective-wise outcomes, aligned with Chapter 1, are summarized below:

1. **Objective 1 (Hybrid CSLR architecture):** A hybrid ResNet-transformer model with joint CTC and cross-entropy supervision was designed and implemented, and it remained stable compared with pure CTC-oriented baselines.
2. **Objective 2 (Evaluation under constrained resources):** Under constrained GPU settings, the model achieved a best test WER of 50.91% on RWTH-PHOENIX-Weather 2014 with reproducible training controls.
3. **Objective 3 (Bidirectional assistive prototype):** A bidirectional pipeline integrating sign-to-text recognition and text-to-sign retrieval was implemented as a functional desktop prototype.
4. **Objective 4 (Usability demonstration):** The multithreaded desktop interface demonstrated functional integration feasibility for interactive assistive communication scenarios.

Overall, each objective defined in Chapter 1 was addressed with implementable system artifacts and measurable outcomes.

7.5 Future Scope

Although the current system is effective as a baseline, several extensions can substantially improve performance:

- **Longer temporal context**: increasing frame coverage beyond 64 frames using memory-efficient training (e.g., gradient checkpointing or mixed precision).
- **Advanced alignment objectives**: incorporating Visual Alignment Constraint (VAC) style losses for stronger monotonic frame-gloss correspondence.
- **Stronger visual encoders**: evaluating larger pretrained video backbones and multi-scale temporal modeling.
- **Error-focused analysis**: systematic gloss-level confusion analysis and signer-wise performance decomposition for targeted improvement.
- **Bidirectional expansion**: improving text-to-sign quality through phrase-level retrieval and smoother transition modeling between clips.

In summary, this work establishes a technically sound and practically deployable foundation for continuous sign language recognition and bidirectional assistive communication systems.

Appendix A

EXPERIMENT EVOLUTION SUMMARY

A.1 Purpose of This Appendix

This appendix summarizes the model families explored during the project using checkpoint metadata and training artifacts available in the repository. The table is included to make the experimentation process auditable and to justify the final model selection in the main text.

A.2 Checkpoint-Based Approach Comparison

Table A.1: Summary of Explored Approaches from Stored Checkpoints

Approach Family	Best Epoch	Recorded Best WER (%)	Notes
Transformer (early baseline)	2	91.28	Unstable alignment behavior
Hybrid (early)	–	–	No stable retained run metric in archived outputs
I3D feature branch	33	94.78	Did not converge to useful quality
End-to-end (e2e) branch	2	95.42	Early checkpoint indicates unstable run
Improved branch	49	60.40	Better than early baselines
ResNet-18 branch	89	51.44	Best archived branch checkpoint (resnet18 best checkpoint file)
VAC family (single implementation, multiple runs)	35	61.43	Same VAC code path; best result across internal runs

Interpretation note: WER values above 100% are possible in sequence tasks when insertion errors are high. This table summarizes archived branch-level checkpoint outcomes and does not replace the final reported thesis result. The final hybrid system performance reported in Chapter 5 is 50.91% test WER. Some checkpoint files store WER as a fraction (for example, 0.5144 $\rightarrow$ 51.44%), while others store percentage directly.

A.3 Implementation-Oriented Approach Matrix

Table A.2 summarizes the approaches that were actually implemented in this repository and aligned with literature-guided design choices.

Table A.2: Implemented Approach Matrix Across Literature-Guided Variants

Approach	**Research-Motivated Design**	**Repository Implementation**	**Experiment Intent**
Transformer baseline	Alignment-focused sequence modeling from CTC-oriented literature	train.py, train_text2gloss.py	Establish baseline behavior and collapse risk profile
Hybrid CTC+Attention	Joint objective inspired by hybrid CTC-attention research	train_hybrid.py	Improve stability and sequence quality under limited resources
End-to-end branch	Unified visual-to-gloss training path for direct pipeline comparison	train_e2e.py	Compare direct end-to-end optimization against modular variants
Improved hybrid	Hybrid training with stronger augmentation and optimization controls	train_improved.py	Reduce overfitting and improve generalization on constrained hardware
I3D branch	Spatiotemporal feature-transfer strategy based on video-model literature	train_i3d.py, i3d model source file	Evaluate whether stronger video features improve CSLR robustness
VAC family	Visual alignment constraint objective for alignment regularization	train_vac.py	Mitigate alignment drift and blank-dominant collapse trends

Scope note: This matrix includes only approaches with explicit code implementations in the repository.

A.4 Suggested Extended Logging for Future Iterations

To strengthen future thesis-grade reproducibility, each run should store a unified experiment card containing:

- complete model hyperparameters,
- optimizer and scheduler settings,
- augmentation flags,
- split-level metrics (dev/test WER, insertion/deletion/substitution rates),

- checkpoint hash and script version.

Such standardized tracking would support cleaner cross-run comparisons and reduce ambiguity in multi-branch experiment narratives.

A.5 Why These Approaches Were Explored

- **Pure/early CTC-transformer baseline:** included to test alignment-only learning behavior motivated by CTC literature.[1]
- **Hybrid CTC-attention objective:** explored based on hybrid objective evidence from sequence modeling literature.[14] The design also followed sign-language transformer work.[7]
- **VAC family:** tested to examine alignment-constraint benefits reported in CSLR.[9]
- **I3D feature branch:** included because inflated 3D ConvNet features have shown strong transfer for video understanding.[15]
- **Improved hybrid branch:** explored to reduce overfitting through enhanced temporal augmentation and gradient accumulation under constrained hardware.

A.6 Reproducibility Artifacts

The following project artifacts support reproducibility of the experiment-evolution narrative:

- Checkpoint folders under `checkpoints/` for each model family.
- Training logs including `training_log_vac.jsonl`.
- Configuration files (where available) under model-specific checkpoint folders.

A.7 Repository Organization and Reproducibility Assets

To make re-execution pathways explicit, the repository structure used in this thesis can be interpreted as follows:

- **Primary training entry points:** `train_hybrid.py`, `train_improved.py`, `train_vac.py`, `train_e2e.py`, and `train_i3d.py`.
- **Evaluation and testing scripts:** `evaluate.py`, `test_model.py`, and diagnostic checks such as `check_system.py`.
- **Core model and loss modules:** `src/models/` and `src/losses/`, including hybrid objective and VAC-related components.
- **Data pipeline components:** `src/data/` and preparation/extraction scripts under `scripts/`.
- **Documentation trail:** technical notes and architecture/training evolution summaries in repository markdown files and thesis chapters.

Operational reproducibility note: run-to-run comparability in this work is based on fixed split usage, explicit training scripts, recorded metrics in JSONL logs, and archived checkpoint-family outcomes. Where a specific artifact is unavailable for a historical branch, the thesis labels that limitation directly instead of inferring missing values.

BIBLIOGRAPHY

[1] Alex Graves, Santiago Fernández, Faustino Gomez, and Jürgen Schmidhuber. Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 369–376, 2006.

[2] Jens Forster, Christoph Schmidt, Thomas Hoyoux, Oscar Koller, Uwe Zelle, Justus Piater, and Hermann Ney. Rwth-phoenix-weather: A large vocabulary sign language recognition and translation corpus. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC)*, pages 3785–3789, 2012.

[3] Amanda Duarte, Shruti Palaskar, Lucas Ventura, Deepti Ghadiyaram, Kenneth DeHaan, Florian Metze, Jordi Torres, and Xavier Giro-i Nieto. How2sign: A large-scale multimodal dataset for continuous american sign language. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2735–2744, 2021.

[4] Bowen Shi, Diane Brentari, Greg Shakhnarovich, and Karen Livescu. Open-domain sign language translation learned from online video. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6365–6379, 2022.

[5] Oscar Koller, Jens Forster, and Hermann Ney. Continuous sign language recognition: Towards large vocabulary statistical recognition systems handling multiple signers. *Computer Vision and Image Understanding*, 141:108–125, 2015.

[6] Necati Cihan Camgöz, Simon Hadfield, Oscar Koller, Hermann Ney, and Richard Bowden. Neural sign language translation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7784–7793, 2018.

[7] Necati Cihan Camgöz, Simon Hadfield, Oscar Koller, Hermann Ney, and Richard Bowden. Sign language transformers: Joint end-to-end sign language recognition and translation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10023–10033, 2020.

[8] Mathieu De Coster, Mieke Van Herreweghe, and Joni Dambre. Sign language recognition with transformer networks. In *Proceedings of The 12th Language Resources and Evaluation Conference (LREC)*, pages 6018–6024, 2020.

[9] Zhuangzhu Zhu, Dongyue Li, and Limin Wang. Visual alignment constraint for continuous sign language recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11542–11551, 2021.

[10] Lianyu Hu, Liqing Gao, Zekang Liu, and Wei Feng. Continuous sign language recognition with correlation network. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1–10, 2023.

[11] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[12] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is space-time attention all you need for video understanding? In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 813–824, 2021.

[13] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. Videomae: Masked autoencoders are data-efficient learners for self-supervised video pre-training. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 10078–10093, 2022.

[14] Shinji Watanabe, Takaaki Hori, Suyoun Kim, John R Hershey, and Tomoki Hayashi. Hybrid ctc/attention architecture for end-to-end speech recognition. *IEEE Journal of Selected Topics in Signal Processing*, 11(8):1240–1253, 2017.

[15] Joao Carreira and Andrew Zisserman. Quo vadis, action recognition? a new model and the kinetics dataset. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6299–6308, 2017.

[16] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[17] Ilya Loshchilov and Frank Hutter. Sgdr: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations (ICLR)*, 2017.

[18] Rafael Müller, Simon Kornblith, and Geoffrey Hinton. When does label smoothing help? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

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